

White Paper – VMS System With Cost-Effective Media Solves Video Storage Problems

Summary

With today's growing emphasis on security and surveillance, many organizations find themselves adding hundreds -even thousands- of video surveillance cameras. Add high-definition resolution capabilities of today's cameras and organizations need a truly cost-effective video surveillance retention solution. Traditional hard disk-based systems become cost-prohibitive too quickly. Video surveillance retention needs are best met by a system that includes a multi-tiered storage infrastructure.

The Problem

Many of today's businesses -such as hospitals, universities, airports, casinos, and others- need a video surveillance system either as a replacement for an aging CCTV system or as a brand new installation. Often this is in response to a growing expectation that video evidence will be available in the event of a crime, accident, or other incident or dispute. In today's society, the first thing a law enforcement agency or legal representative of either party in a dispute will ask for is a copy of any available video footage of the incident in question. The ability to quickly and easily provide high quality video evidence of the incident in question will be a big help to law enforcement agencies in narrowing down suspects in case of a crime.

The same sort of positive reliable evidence can limit the liability of an organization in case of a lawsuit regarding an accident by providing undisputable proof of the parties involved, the timeline, and the sequence of events that took place. In other cases, the need for a video surveillance system may be driven by regulatory requirements, such as in the health care industry. These requirements can be national, statewide, or local.

However, the high-cost of video storage retention significantly reduces the length of time -or even the quality of the recorded video- the company can afford to save. This directly impacts the overall liability exposure of the company. To keep the quality of video for the longest period of time desired, organizations today need a flexible, cost effective, means of capturing video evidence and saving it in easily searchable and retrievable form for as long as it is needed.

The Solution

Item 1 – The Cameras

Video camera technology has come a long way, even in the past few years. Driven by the demand for high quality, low cost cameras in cell phones and tablets, enormous advances have been made in both the quality and cost of digital video capture devices.

Today's cameras are capable of providing extremely high resolution, frame rates of 30 frames per second and higher, full color video in anything from bright sunlight to very low-level light. One can also purchase cameras with built-in infrared emitters if night time surveillance is desirable or required.

For the absolute best quality video with stunning detail vendors have introduced 8K (37megapixel) surveillance cameras at affordable prices.

An inhibitor to utilizing these high resolution, high frame rate, full color surveillance cameras is the exorbitant amount of storage space they consume.

Item 2 – The Hard Disk Drives

Almost all video surveillance systems on the market today utilize hard disk drive-based storage systems. The underlying concept is that hard drives are cheap. Although generic hard drives are indeed inexpensive, hard drives that are specifically manufactured for constant 24/7/365 writing of video data, from vendors like Western Digital and Seagate cost more.

Then you have to place those hard drives into some sort of hardware device to manage all of those individual drives as a complete storage system. There are a number of systems available for such use: NAS, SAN, JBOD, NVR, are the most common. And almost all of those devices also implement some form of RAID solution, which helps protect the survivability of the data at the cost of reducing the amount of available video storage capacity.

It is beyond the scope of this white paper to describe and explain each of those types of storage systems or the RAID implementations that are used. For the purpose of this white paper and our storage calculations we will assume a NAS with RAID6.

Vendors and solution providers have been delivering various combinations of these systems since the introduction of IP-based video solutions. What has become a commonplace ideology is that at a certain point of video storage requirement, "you just can't afford it."

And we really have not discussed how hard drives can crash and fail (thus the need to implement a RAID capability).

Item 3 – The Work-Arounds

The physical security industry immediately tried to find ways to store large amounts of video without taking up so much disk space. Concepts such as H.264, H.265, and record-on-motion are most popular.

The introduction of H.264 and H.265 has significantly helped in reducing the amount of needed storage for video feeds. This video algorithm has the ability to maintain the level of quality video while saving a hefty percentage of storage space.

The challenge with H.264/H.265 is that as the number of actual cameras increases within any organization, the need for more and more storage space still arises and we are back to our vicious cycle.

Vendors also introduced the concept of “Record-on-Motion” through which the software would only store video “where there is activity.” But then we had issues with trees moving in the background or birds flying through the scene, or lights flickering that cause “motion” but didn’t have any “real” activity. So, the vendors allowed for “thresholds” of activity to be ignored. All this did was cause systems to NOT record and store video when something important DID happen. But at least it saved storage space.

Although overall helpful, these work-arounds have been focused on ways to save hard disk storage space.

Item 4 – Calculating Storage Needs

Let’s walk through the math to understand how much storage space is needed for video surveillance retention needs. (As an interesting research project, Google “video storage calculator” and try out a number of those calculators with the exact same ‘inputs’ and see the different results generated.)

First, we need to know the resolution that our video surveillance camera is recording at. This is represented in pixels. Let’s use a “1080p” IP-camera resolution. This could be 1920 pixels by 1080 pixels. When you multiply the height by the width you will get 2,073,600 pixels. This would be for each frame of video. If you record at full motion video, that is 30-frames-per-second.

(NOTE: Directly in response to the storage challenge, individuals will argue that you don’t need to record at 30-frames-per-second. But if the camera can output at 30-frames-a-second -or even higher- why would you not want to view at that rate? [Answer: Storage requirement limitations])

Take the single frame resolution and multiple by the number of frames per second. Turn those pixels into bits (multiply 24-bits for color or 2-bits for black and white). Turn that number into kilobits per second. The math gets a little difficult here as you need to multiply the ‘motion’ of the scene if you are using the H.264/H.265 compression protocol. Cozaint’s own video storage calculator indicates that at 40% motion you would generate a bitrate of 6,221-kilobits per second (kps). For one camera.

If we take the KPS and turn them into how much storage it would use per day we get 64.07-gigabytes per day. For one camera.

Now how many cameras do you want? And how many days of retention to you want? Multiply all those numbers. Let's say we want 90-days of retention for 400 cameras. Our calculator says that is 2,252-terabytes of storage required. Stated another way, that is 2.25-petabytes of storage.

You can see how this storage requirement gets quite large, quite fast.

If we look at the total cost of ownership (TCO) for 2.25-petabytes, we see that this is a large number of hard drives, the hardware needed for all those hard drives, and the amount of extra hard drives needed for the RAID implementation. Add in the electricity needed to keep that many hard drives spinning, the cooling costs needed to keep the hardware within proper operating temperatures and you get an expensive proposition.

Now, you see why so many operators will turn down the resolution of their cameras; turn down the frame rates; turn on 'record only on motion'; and use other idea to reduce storage space.

Is there any other way we can use the best quality resolution, at the best frame rates, with the highest color quality AND still not spend a fortune on storage equipment?

Item 5 – What is LTO?

Linear Tape Open (LTO) Ultrium is a high-capacity, single-reel data tape storage solution developed and continually enhanced by Hewlett Packard Enterprise, IBM, and Quantum and promoted by the LTO Program.

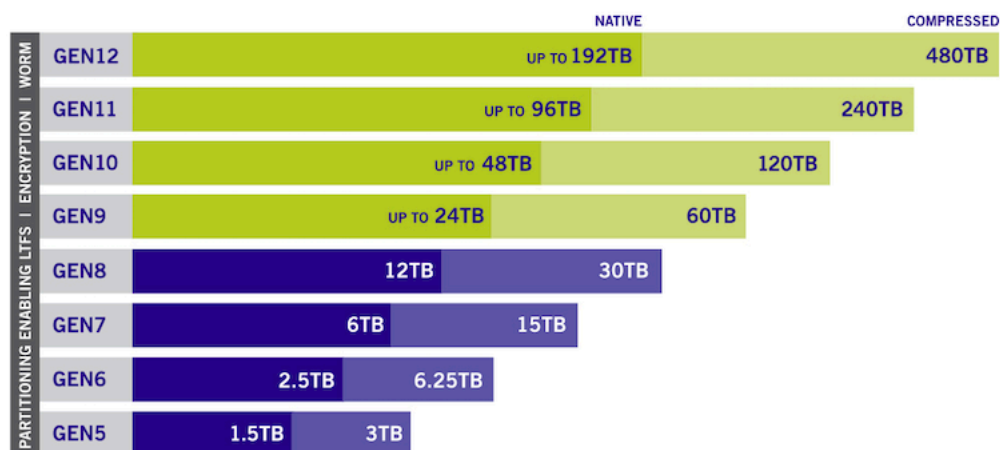
First introduced in 2000, LTO technology is currently in its 8th generation. LTO-8 specifications support each cartridge with capacity of 12-terabytes and data transfer rates of up to 300MB per second for over 1-terabytes of storage performance an hour per drive.

If hard disk-based storage solutions become very expensive in large capacities, what other storage technologies exist that could scale to such capacities?

LTO is a powerful, scalable, and adaptable data tape format that helps address the growing demands of long-term video retention.

LTO ULTRIUM ROADMAP

ADDRESSING YOUR STORAGE NEEDS



LTO Data Tape remains unrivaled in terms of cost for capacity, reliability, portability and security and it continues to play a crucial role in data protection. LTO perfectly complements a disk-based storage system by providing a nearline storage offering.

Add LTFS. The Linear Tape File System (LTFS) makes viewing and accessing tape files easier than ever. LTFS works in conjunction with LTO technology for ease of use and portability for open systems data tape storage. With LTFS, one partition holds the content and the other holds the content's index, so the tape cartridge can be self-describing to improve archive management.

Putting It All Together – *askALICE* From Cozaint

An ideal video storage infrastructure would provide medium- and long-term storage and:

- It should provide easy access to all stored video
- Video quality should never degrade over time
- There should be no 'extra' steps necessary to playback any/all recorded video
- It should easily scale as I need to add surveillance cameras

If we look to other industries that have a similar need for long-term video retention, we can look at the Media and Entertainment industry (think 'Hollywood': See: [Hollywood comes to Video Surveillance LINK](#)). The M&E market has long been using LTO as a way to move content from hard disk-based systems to the long-term (read: decades!) reliable storage.

The ideal solution is a combination of disk-based storage and LTO-based storage. In such a multi-tiered storage infrastructure, a much smaller amount of hard disk storage is used as an initial storage tier with the more cost effective LTO storage tier utilized for complete and long-term retention.

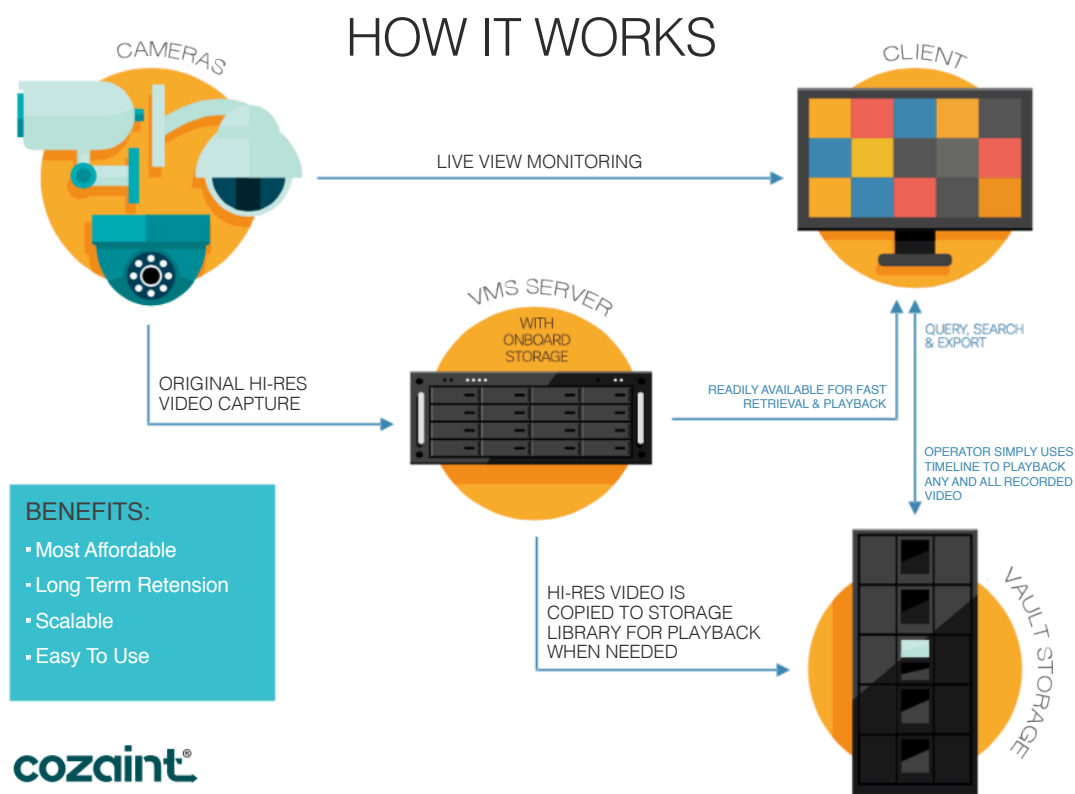
If we use the 2.5-petabytes example above we could build out an infrastructure in the following manner:

- a) 10% HDD = 250 TB
- b) LTO library with 375 slots and four (4) drives
- c) Such a complete solution would cost: \$131,800

If we went with an all hard disk based solution it would need:

- a) Over 225 hard drives
- b) The equipment to management all those hard drives
- c) Such a complete hard drive solution would cost: \$326,800*

This is more than a 40% savings.



Cozaint has taken the requirements of such an affordable network video recording system and put together the appropriate software and hardware pieces to fulfill the needs of most medium to large scale video surveillance installations. Combining specialized off the shelf PC hardware running the innovative **BOBBYvms** surveillance software and LTO libraries, Cozaint has come up with the most economical, flexible, scalable long-term video retention solutions on the market.

The **BOBBYvms** software comes with modules to add cameras, set up and modify user defined rules to direct the flow of video data, observe and control any camera in the system, maintain the video storage, and track the flow of video to the LTO library.

Cozaint's **askALICE** solution affordably scales from hundreds of terabytes to petabytes of video retention. Let us show you how we can deliver the long-term video storage you need at costs significantly lower than any other solution.

About Cozaint Corp

Founded in 2018, Cozaint Corporation is a service-disabled veteran owned business that manufactures and develops 'smart' physical security solutions. Physical security towers and kiosks and video surveillance storage systems that deliver 24/7/365 surveillance and peace-of-mind. Our tower and kiosk units are connected to our live care center for concierge assistance (and panic connection). Learn more at www.Cozaint.com

** Using Western Digital Purple surveillance hard drives in a NAS storage solution.*